

A CMS Open Data Diagnostic Search for H to Tau Tau in the Mu Tau_h Final State

Analysis my_analysis

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Abstract

A diagnostic search for H to tau tau is performed with CMS 2012 Open Data in the mu tau_h final state using reduced Run2012B/C TauPlusX samples. A binned profile-likelihood fit to classifier-score templates in VBF, boosted, and zero-jet categories gives $\mu_{\text{hat}} = 7.664$, an observed 95% CLs upper limit $\mu < 13.323$, and an observed q_0 diagnostic $Z = 2.978$. The score-template validation is flagged in both the 10% validation sample and the full data, so these values are simplified open-data diagnostics and are not CMS-quality evidence for H to tau tau.

1 Introduction

The H to tau tau decay probes the Higgs coupling to charged leptons. CMS reported Run 1 evidence for this decay and later observed it with 13 TeV data (CMS Collaboration 2014, 2018). This paper draft presents a compact open-data diagnostic using the public reduced CMS 2012 H to tau tau samples (CMS Collaboration 2021a). The analysis is limited to the mu tau_h final state and is intended as a reproducible public-data workflow, not a replacement for the full CMS publications.

2 Data And Simulation

The data are Run2012B and Run2012C TauPlusX reduced NanoAOD-like samples from the CERN Open Data Portal (CMS Collaboration 2021d, 2021e). The integrated luminosity is 11.467/fb, following the CMS Open Data H to tau tau tutorial and 2012 luminosity provenance (CMS Open Data Analyses 2021; CMS Collaboration 2017). The luminosity uncertainty is 2.6% (CMS Collaboration 2013).

Sample group	Samples	Use
Data	Run2012B/C TauPlusX	observed signal and control regions
Signal	ggH and VBF H to tau tau	fitted signal templates
Background	DYJetsToLL, TTbar, W1/W2/W3JetsToLNu	fitted background templates
Unavailable in reduced set	QCD, diboson, single top, embedded Z to tau tau, W4/inclusive W	documented limitation

The signal and background records are the public CMS Open Data reduced samples (CMS Collaboration 2021c, 2021g, 2021b, 2021f, 2021h, 2021i, 2021j). MC normalization uses official Open Data event denominators and cited cross-section inputs; it is not derived by matching data yields.

3 Event Selection And Categories

Events are required to pass HLT_IsoMu17_eta2p1_LooseIsoPFTau20. Muons have $pt > 20$ GeV and $|\eta| < 2.1$ with tight identification and isolation. Hadronic-tau candidates have $pt > 20$ GeV, $|\eta| < 2.3$, decay-mode ID, tight isolation ID, and tight anti-muon ID. The signal-region selection requires an opposite-sign mu tau_h pair and muon-MET transverse mass below 40 GeV.

The fit uses three mutually exclusive categories: VBF, boosted, and zero-jet. VBF events have at least two clean jets with leading-dijet mass above 300 GeV and leading-dijet $|\Delta\eta|$ above 2.5. Boosted events have at least one clean jet after failing the VBF selection, and zero-jet events have no clean jets.

Category	Data	Background	Signal	Data/background
VBF	79	146.67	1.590	0.539
Boosted	2213	2062.80	9.175	1.073
Zero-jet	8466	6894.80	14.341	1.228

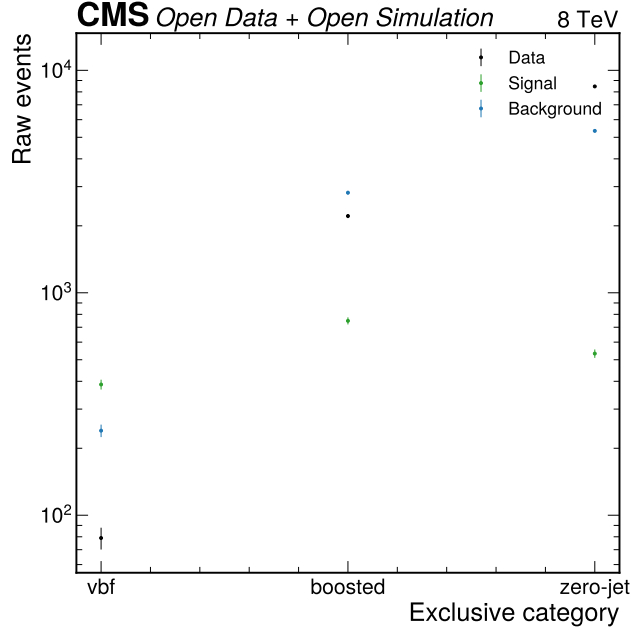


Figure 1: Selected category yields. The figure shows the event yields in the three exclusive fit categories for data, signal MC, and background MC. It documents the channel composition of the simultaneous likelihood fit and shows that the zero-jet channel dominates the selected event count.

4 Statistical Analysis

The fitted observable is a histogram-gradient-boosting classifier score in the three event categories. The likelihood is a binned profile likelihood implemented with pyhf (Heinrich et al. 2021),

$$L(\mu, \theta) = \prod_{c,b} \text{Pois}(n_{cb} | \mu s_{cb}(\theta) + b_{cb}(\theta)) \prod_k \pi_k(\theta_k). \quad (1)$$

Upper limits use the CLs construction (Read 2002), and the discovery diagnostic uses the one-sided q_0 asymptotic approximation (Cowan et al. 2011). The W+jets rate is constrained by a high-mT control-region scale 0.8528 ± 0.0370 . The retained nuisances include luminosity, DY normalization, tau open-data acceptance, W high-mT normalization, and MC statistical uncertainties.

5 Results

The expected median 95% CLs upper limit is $\mu < 4.199$, and the expected discovery diagnostic is $Z = 0.526$. The full-data fit gives the diagnostic results summarized in Table 1.

Table 1: Main open-data fit results. The observed entries are not CMS-quality evidence because score-template validation is flagged.

Quantity	Value	Status
Expected median 95% CLs limit	$\mu < 4.199$	background Asimov
Expected discovery diagnostic	0.526	background Asimov
Observed μ_{hat}	7.664	flagged diagnostic
Observed 95% CLs limit	$\mu < 13.323$	flagged diagnostic
Observed $q_0 Z$	2.978	flagged diagnostic

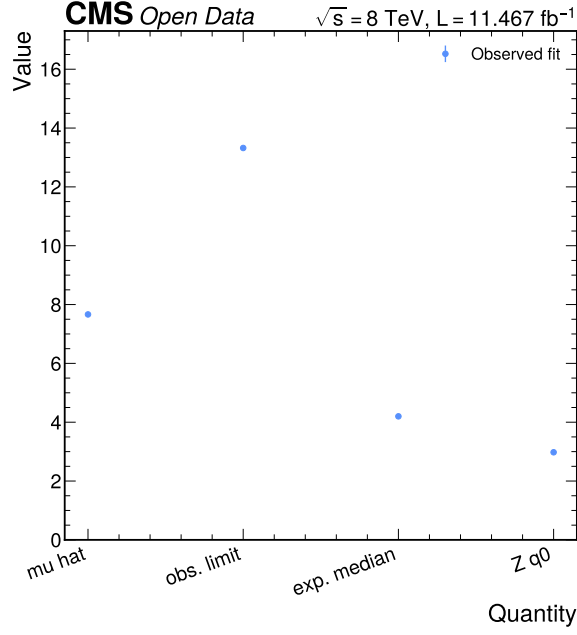


Figure 2: Observed result summary. The figure summarizes the fitted signal strength, observed upper limit, expected median limit, and observed q0 diagnostic. The values are diagnostics of the reduced open-data model and must be read together with the validation caveat.

6 Validation Caveat

The score-template model is not validated at CMS-publication quality. In the 10% validation sample, the combined data/background ratio is 1.232 with χ^2/ndf 1.732 and status **flagged**. In the full data, the combined data/background ratio is 1.182 with χ^2/ndf 3.218 and status **flagged**. Therefore the observed $Z = 2.978$ and $\mu_{\text{hat}} = 7.664$ are simplified open-data diagnostics, not evidence comparable to CMS publications.

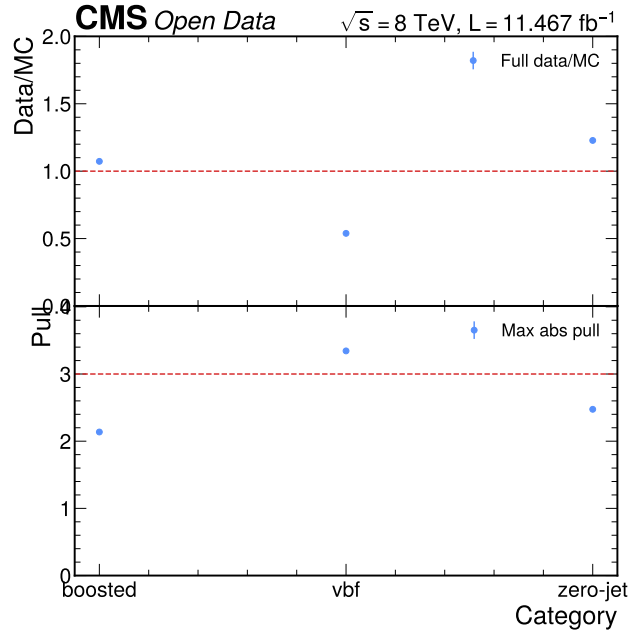


Figure 3: Full-data score validation summary. The figure summarizes data/background ratios and maximum pulls in the observed score templates. It is the main diagnostic showing why the observed fit result is not promoted to a CMS-quality evidence statement.

7 Comparison With Published Results

The open-data result is compared with CMS H to tau tau publications only as context. CMS Run 1 used 7 and 8 TeV data with multiple channels and reported $\mu = 0.78 \pm 0.27$ with local significance above three standard deviations (CMS Collaboration 2014). CMS 2018 used 13 TeV data and reported observed and expected significances of 4.9 and 4.7 with $\mu = 1.09$ and roughly 0.27 uncertainty (CMS Collaboration 2018). These publication results are not direct pass/fail targets for the reduced single-channel open-data model.

Table 2: Comparison of the open-data diagnostic with published CMS H to tau tau results. The entries differ in energy, channel coverage, calibration completeness, and validation quality.

Result	Scope	Significance	Signal-strength information
This open-data expected	8 TeV mu tau_h reduced samples	0.526	expected $\mu < 4.199$
This open-data observed	8 TeV mu tau_h reduced samples	2.978 flagged	$\mu_{\text{hat}} = 7.664$; $\mu < 13.323$
CMS Run 1	7+8 TeV multi-channel	>3	$\mu = 0.78 \pm 0.27$
CMS 2018	13 TeV multi-channel	observed 4.9, expected 4.7	$\mu = 1.09 \pm \text{about } 0.27$

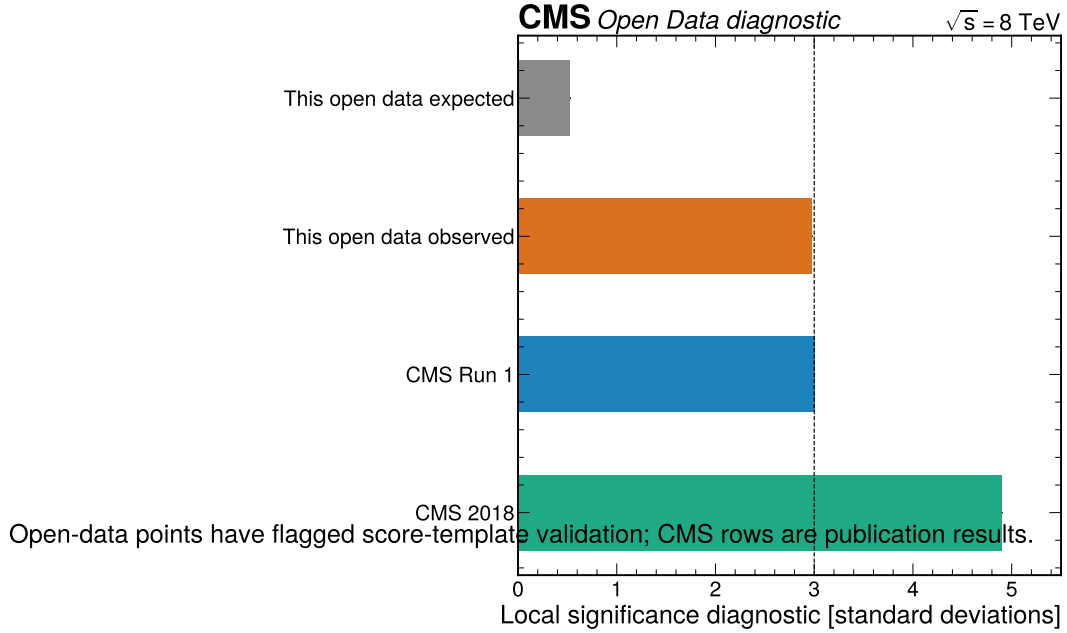


Figure 4: Significance comparison. The figure compares the open-data expected and observed diagnostic significances with CMS publication values. The open-data observed point is explicitly labelled as flagged and should not be interpreted as CMS-quality evidence.

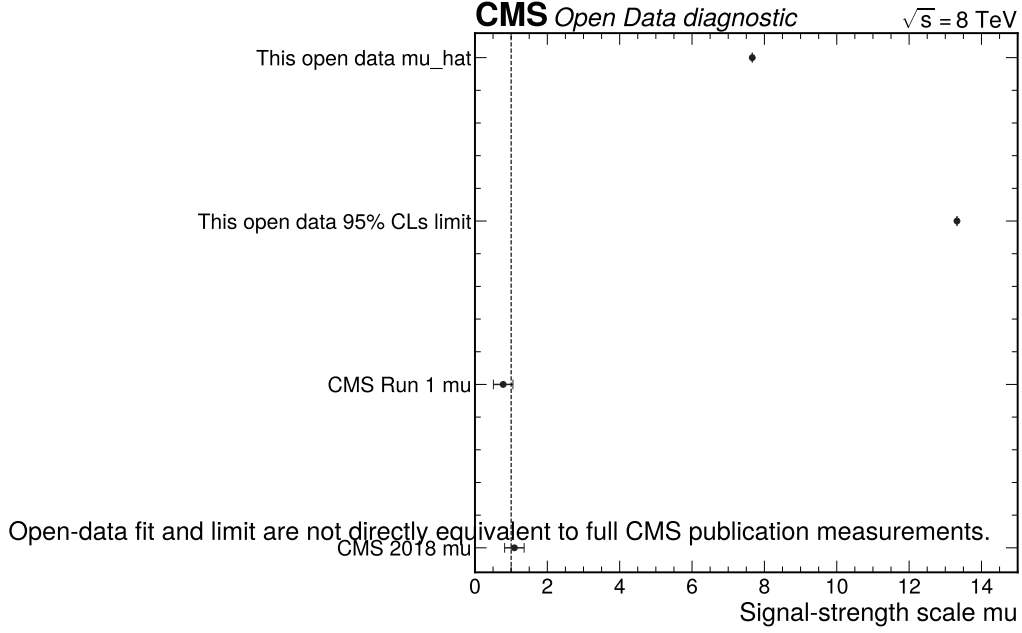


Figure 5: Signal-strength and limit comparison. The figure compares open-data signal-strength-scale diagnostics with CMS Run 1 and CMS 2018 signal-strength measurements. The open-data limit and μ_{hat} are not directly equivalent to the published measurements because the reduced public model has flagged score-template validation.

8 Conclusion

A reproducible reduced CMS 2012 Open Data H to tau tau diagnostic search has been documented in the $\mu_{\text{tau_h}}$ final state. The full-data fit gives $\mu_{\text{hat}} = 7.664$, $\mu < 13.323$ at 95% CLs, and $Z = 2.978$. Because the 10% and full-data score-template validations are flagged, these are simplified open-data diagnostics and not CMS-quality evidence. The analysis is useful as a public workflow and as a scoped comparison to CMS H to tau tau publications, not as an independent publication measurement.

Reproducibility Note

The markdown, TeX, PDF, and comparison figures are generated from the committed Phase 4 JSON outputs by `phase5_documentation/src/build_phase5_docs.py`.

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